

Appendix C: Motor Drive Sizing

Appendix C: Motor and Drive Sizing

C.1 Stepper Motor Specifications

C.1.1 NEMA Stepper Motor Frame Sizes

NEMA Size	Flange (mm)	Length Range (mm)	Shaft Dia (mm)	Typical Torque Range (N*m)	Applications
NEMA 11	28 x 28	30-51	5	0.08-0.15	Small 3D printers, light actuators
NEMA 14	35 x 35	30-48	5	0.15-0.30	Compact CNC, extruders
NEMA 17	42 x 42	34-60	5	0.26-0.65	3D printers, desktop CNC, pick-place
NEMA 23	57 x 57	56-112	6.35 or 8	0.9-3.0	Mid-size CNC, plasma tables, routers
NEMA 34	86 x 86	80-156	12.7 or 14	4.0-12.0	Heavy CNC, large gantry systems
NEMA 42	110 x 110	114-201	19	12.0-30.0	Industrial CNC, high-inertia loads

Note: NEMA = National Electrical Manufacturers Association. Number indicates mounting hole spacing in inches x 10 (NEMA 23 = 2.3" spacing).

C.1.2 Stepper Motor Wiring Configurations

Bipolar (4-wire): - One winding per phase (2 phases total) - Requires H-bridge driver (reverses current direction) - Higher torque than unipolar (full coil energized) - Most common for CNC applications

Unipolar (6-wire or 8-wire): - Center-tapped windings - Simpler driver (just switches, no H-bridge) - Lower torque (only half coil energized at a time) - Obsolete for modern CNC (bipolar preferred)

Bipolar-Series vs. Bipolar-Parallel (8-wire motors): - **Series:** Higher inductance -> better low-speed torque, lower top speed - **Parallel:** Lower inductance -> better high-speed torque, higher top speed - **Rule of thumb:** Series for direct drive, parallel for 2:1+ reduction gearboxes

C.1.3 Stepper Motor Torque-Speed Curves

Key Parameters: - **Holding Torque (T_H):** Maximum torque at 0 RPM (standstill) - **Pull-out Torque (T_{PO}):** Maximum torque before stalling at given speed - **Pull-in Torque (T_{PI}):** Maximum starting torque without ramping - **Resonance Region:** 100-200 full steps/sec typical (causes vibration/stalling)

Torque vs. Speed Relationship:

At low speeds (0-500 RPM):

$$T \approx T_H \left(1 - \frac{\omega}{3000}\right)$$

At high speeds (500+ RPM):

$$T \approx \frac{K_v \cdot V_{supply}}{\omega}$$

where: - T = available torque (Nm) - T_H = holding torque (Nm) - ω = rotor speed (RPM) - K_v = motor voltage constant - V_{supply} = driver supply voltage (V)

Example Torque Curve (NEMA 23, 3.0 N*m holding torque, 48V supply):

Speed (RPM)	Available Torque (N*m)	% of Holding Torque
0	3.00	100%
100	2.85	95%
300	2.50	83%
600	1.80	60%
1000	1.20	40%
1500	0.75	25%
2000	0.50	17%
3000	0.25	8%

Increasing High-Speed Torque: 1. **Increase supply voltage** (most effective): 48V -> 72V typically doubles torque at 1500 RPM 2. **Use low-inductance motors:** Reduces L/R time constant, faster current rise 3. **Microstepping with higher current:** 1/8 or 1/16 microstepping smooths motion 4. **Gearbox reduction:** Trade speed for torque (2:1 reduction = 2x torque, 0.5x speed)

C.2 Servo Motor Specifications

C.2.1 Servo vs. Stepper Comparison

Characteristic	Stepper Motor	Servo Motor
Control type	Open-loop (no feedback)	Closed-loop (encoder feedback)
Positioning accuracy	+/-0.05 deg (1/4 step) typical	+/-0.001 deg with high-res encoder
Torque at high speed	Poor (drops rapidly)	Excellent (flat to rated speed)
Holding torque	Full torque at standstill	Zero torque at standstill (unless energized)

Characteristic	Stepper Motor	Servo Motor
Overload behavior	Stalls (loses position)	Faults/shuts down (maintains position tracking)
Efficiency	40-60% (always energized)	80-95% (energized on demand)
Cost	Low cost	—\$
Noise	Moderate (resonance)	Low (smooth motion)
Best for	Low cost, open-loop OK	High performance, critical positioning

C.2.2 AC Servo Motor Frame Sizes (Standard)

Frame Size (mm)	Rated Power (W)	Cont. Torque (N*m)	Peak Torque (N*m)	Rated Speed (RPM)	Applications
40	50-100	0.16-0.32	0.48-0.95	3000	Ultra-compact automation
60	100-400	0.32-1.27	0.95-3.82	3000	Pick-place, light CNC
80	400-750	1.27-2.39	3.82-7.16	3000	Small/mid CNC axes
110	750-1500	2.39-4.77	7.16-14.3	3000	Mid-size CNC, mill axes
130	1500-2000	4.77-6.37	14.3-19.1	3000	Large CNC, high-load axes
180	3000-5000	9.55-15.9	28.6-47.7	3000	Heavy industrial CNC

Continuous vs. Peak Torque: - **Continuous (rated) torque:** Can run indefinitely without overheating
- **Peak torque:** Available for short bursts (1-2 seconds typical), acceleration/deceleration - **Duty cycle:** Peak torque limited by thermal constraints (motor winding temperature)

Encoder Resolution: - **Standard:** 2500 PPR (pulses per revolution) -> 10,000 counts/rev with quadrature
- **High-resolution:** 10,000-20,000 PPR -> 40,000-80,000 counts/rev - **Absolute encoders:** Retain position when powered off (battery-backed or multi-turn)

C.2.3 Servo Motor Torque-Speed Curves

Typical AC Servo Characteristics: - Flat torque from 0 to rated speed (3000 RPM typical) - Constant power above rated speed (torque inversely proportional to speed) - Peak torque = 3x continuous torque (thermal limit)

Torque Curve Regions:

$$T(\omega) = \begin{cases} T_{cont} & \omega \leq \omega_{rated} \text{ (constant torque region)} \\ \frac{P_{rated}}{\omega} & \omega > \omega_{rated} \text{ (constant power region)} \end{cases}$$

where: - T_{cont} = continuous rated torque (N*m) - ω_{rated} = rated speed (rad/s or RPM) - P_{rated} = rated power (W)

Example: 1.5 kW, 4.77 N*m servo (rated speed 3000 RPM):

Speed (RPM)	Continuous Torque (N*m)	Peak Torque (N*m)	Power (W)
0	4.77	14.3	0
1000	4.77	14.3	500
3000	4.77 (rated)	14.3	1500 (rated)
4500	3.18	9.5	1500
6000	2.39	7.2	1500

C.3 Motor Sizing Calculations

C.3.1 Required Torque Calculation

Linear Axis (Ball Screw Drive):

$$T_{motor} = \frac{F \cdot p}{2\pi\eta} + T_{friction} + T_{inertia}$$

where: - T_{motor} = motor torque required (N*m) - F = cutting force or load force (N) - p = ball screw lead (m/rev) - η = efficiency (0.90-0.95 for ball screws) - $T_{friction}$ = friction torque (guide rails, seals) - $T_{inertia}$ = torque to accelerate load

Inertia Torque (Acceleration):

$$T_{inertia} = J_{total} \cdot \alpha$$

where: - J_{total} = total reflected inertia (kg*m²) - α = angular acceleration (rad/s²)

Reflected Inertia (Linear Load to Motor Shaft):

$$J_{reflected} = m \cdot \left(\frac{p}{2\pi}\right)^2$$

where: - m = moving mass (kg) - p = screw lead (m/rev)

Example: NEMA 23 stepper sizing for plasma table X-axis

Given: - Moving mass: $m = 15$ kg (gantry assembly) - Ball screw lead: $p = 0.010$ m/rev (10mm/rev, RM2005) - Cutting force: $F = 50$ N (plasma drag force) - Max velocity: $v = 200$ mm/s = 0.2 m/s - Acceleration time: $t = 0.5$ s - Efficiency: $\eta = 0.90$

Step 1: Screw speed at max velocity

$$\omega = \frac{v}{p} = \frac{0.2}{0.010} = 20 \text{ rev/s} = 1200 \text{ RPM}$$

Step 2: Reflected inertia

$$J_{reflected} = 15 \times \left(\frac{0.010}{2\pi}\right)^2 = 3.8 \times 10^{-5} \text{ kg*m}^2$$

Motor rotor inertia (NEMA 23): $J_{motor} = 1.2 \times 10^{-4} \text{ kg*m}^2$

$$J_{total} = J_{motor} + J_{reflected} = 1.58 \times 10^{-4} \text{ kg} \cdot \text{m}^2$$

Step 3: Angular acceleration

$$\alpha = \frac{\omega}{t} = \frac{20 \times 2\pi}{0.5} = 251 \text{ rad/s}^2$$

Step 4: Torque components

Load torque:

$$T_{load} = \frac{50 \times 0.010}{2\pi \times 0.90} = 0.088 \text{ N} \cdot \text{m}$$

Inertia torque:

$$T_{inertia} = 1.58 \times 10^{-4} \times 251 = 0.040 \text{ N} \cdot \text{m}$$

Friction torque (estimate 10% of load):

$$T_{friction} = 0.010 \text{ N} \cdot \text{m}$$

Total torque required:

$$T_{required} = 0.088 + 0.040 + 0.010 = 0.138 \text{ N} \cdot \text{m}$$

At 1200 RPM, NEMA 23 (3.0 Nm holding torque, 48V) provides ~1.2 Nm -> 8.7x safety factor (adequate)

C.3.2 Motor Sizing for Rotary Axis (4th Axis, Spindle Indexer)

Torque Required:

$$T = J_{load} \cdot \alpha + T_{friction} + T_{cutting}$$

Load Inertia (Solid Cylinder):

$$J = \frac{1}{2}mr^2$$

Example: Rotary table for indexing

Given: - Chuck + part mass: $m = 10 \text{ kg}$ - Effective radius: $r = 0.1 \text{ m}$ - Acceleration time: $t = 1 \text{ s}$ to 60 RPM - Cutting torque: $T_{cutting} = 5 \text{ N} \cdot \text{m}$ (milling force x radius)

Load inertia:

$$J_{load} = \frac{1}{2} \times 10 \times 0.1^2 = 0.05 \text{ kg} \cdot \text{m}^2$$

Angular acceleration:

$$\alpha = \frac{60 \times 2\pi/60}{1} = 6.28 \text{ rad/s}^2$$

Inertia torque:

$$T_{inertia} = 0.05 \times 6.28 = 0.314 \text{ N} \cdot \text{m}$$

Friction (bearing + worm gear, assume 20%):

$$T_{friction} = 1.0 \text{ N} \cdot \text{m}$$

Total:

$$T_{required} = 0.314 + 1.0 + 5.0 = 6.3 \text{ N}\cdot\text{m}$$

With 10:1 worm gear (eta = 0.50 efficiency):

$$T_{motor} = \frac{6.3}{10 \times 0.50} = 1.26 \text{ N}\cdot\text{m}$$

****NEMA 23 with 3.0 N*m holding torque provides 2.4x margin****

C.4 Stepper Motor Driver Selection

C.4.1 Driver Voltage and Current Ratings

Voltage Selection: - **24V:** Small steppers (NEMA 17), low speed (<500 RPM) - **48V:** NEMA 23, medium speed (1000-1500 RPM) - **most common for CNC** - **72-80V:** NEMA 34, high speed (2000+ RPM), high-inductance motors

Rule of Thumb: Driver voltage = 20-30x motor rated voltage

Example: Motor rated 3.0V, 3.0A -> use 48-72V driver

Current Rating: Driver continuous current $\geq 1.2x$ motor rated current (for microstepping overhead)

C.4.2 Microstepping Settings

Microstep Resolution	Steps/Rev (1.8 deg motor)	Resolution	Smoothness	Torque Loss	Typical Use
Full Step	200	Lowest	Rough, resonance	0%	Obsolete for CNC
Half Step	400	Low	Better	~5%	Low-precision applications
1/4 Step	800	Medium	Good	~10%	General CNC (direct drive)
1/8 Step	1600	High	Smooth	~15%	Standard CNC setting
1/16 Step	3200	Very High	Very smooth	~20%	High precision, low speed
1/32 Step	6400	Ultra High	Smoothest	~25%	Ultra-precision, very low speed

Recommendation: - **1/8 microstepping** for most CNC (good balance of resolution and torque) - **1/16 or 1/32** for very low-speed applications (Z-axis fine positioning) - **1/4 microstepping** for high-speed/high-torque axes (rapids)

C.4.3 Common Stepper Drivers

Driver Model	Voltage Range	Current/Axis	Axes	Microstepping	Interface	Price
DM542	20-50V DC	1.0-4.2A	1	1/2 to 1/128	Step/Dir	Low cost
DM860H	24-80V DC	2.4-7.2A	1	1/2 to 1/128	Step/Dir	Medium cost
Leadshine DM856	24-80V AC	2.0-5.6A	1	1/2 to 1/128	Step/Dir + RS232	Medium cost
Gecko G203V	18-80V DC	0-7.0A	1	1/10 (unique)	Step/Dir	High cost
ClearPath SDSK	48-75V DC	13A peak	1	Servo (encoder)	Step/Dir	Very high cost

C.5 Servo Drive Selection

C.5.1 Servo Drive Types

Pulse/Direction Input (Step/Dir): - Acts like stepper driver (accepts pulse train from motion controller) - Closed-loop internally (encoder feedback to drive) - Easiest retrofit for stepper-based systems - Limited tuning capability

Analog Velocity Input (+/-10V): - Voltage proportional to speed - Common on older CNC controls - Requires separate controller for position loop

Digital Fieldbus (EtherCAT, CANopen, Modbus): - High-speed networked communication - Advanced tuning and diagnostics - Synchronized multi-axis motion - Required for complex path planning

C.5.2 Servo Drive Sizing

Drive Continuous Current: Must meet or exceed motor continuous current rating.

Drive Peak Current: Must provide motor peak current for acceleration/deceleration (typically 3x continuous).

Bus Voltage: Common DC bus voltages: - **48V DC:** Small servos (<400W) - **300-340V DC:** Standard (200-240V AC rectified) - **most common** - **550-680V DC:** High power (380-480V AC rectified)

Regenerative Braking: - Energy returned during deceleration charges DC bus - **Brake resistor** dissipates excess energy if bus voltage rises too high - Size resistor for duty cycle: $P_{brake} = \frac{1}{2}J\omega^2/t_{decel}$ averaged over cycle time

C.5.3 Servo Tuning Parameters

PID Gains (Position Loop): - K_p (**Proportional**): Higher = stiffer, faster response, more oscillation/overshoot - K_i (**Integral**): Eliminates steady-state error, can cause instability if too high - K_d (**Derivative**): Dampens oscillation, reduces overshoot

Typical starting values (requires tuning per application): - $K_p = 100$ (position gain) - $K_i = 5$ (eliminate droop under load) - $K_d = 10$ (damping)

Tuning procedure (Ziegler-Nichols method): 1. Set $K_i = 0$, $K_d = 0$ 2. Increase K_p until sustained oscillation occurs -> note critical gain K_c 3. Set $K_p = 0.6 \times K_c$ 4. Set $K_i = 1.2 \times K_c/T_c$ (where T_c = oscillation period) 5. Set $K_d = 0.075 \times K_c \times T_c$ 6. Fine-tune by observing step response

C.6 VFD (Variable Frequency Drive) for Spindle Motors

C.6.1 VFD Sizing

Power Rating: VFD continuous kW $\geq$ motor nameplate kW

Overload Capacity: - **110% continuous:** General purpose - **150% for 60s:** Heavy duty (frequent start/stop, high inertia loads)

Example: 2.2 kW spindle motor $\rightarrow$ minimum 2.2 kW VFD, recommend 3.0 kW VFD for margin

C.6.2 VFD Control Modes

Mode	Description	Best For	Torque at Low Speed
V/Hz (Scalar)	Voltage proportional to frequency	Constant-torque loads, simple spindles	Poor (<30 Hz)
Sensorless Vector	Estimates rotor position/flux	Improved low-speed torque, variable loads	Good (>5 Hz)
Closed-Loop Vector	Encoder feedback	Constant torque to 0 Hz, precision speed	Excellent (0 Hz)

CNC Spindle Recommendation: Sensorless vector (good performance, no encoder cost)

C.6.3 VFD Input/Output Parameters

Input: - Single-phase 230V AC (up to 2.2 kW typical) - Three-phase 230V or 400V AC (3 kW and above)
- **Derating:** 3-phase VFD run on single-phase input must be derated 50% (e.g., 4 kW VFD $\rightarrow$ 2 kW max on 1-phase input)

Output: - Three-phase AC, variable frequency (0-400 Hz typical) - PWM waveform (carrier frequency 4-16 kHz adjustable) - **Do NOT use motor overload relays on output** (VFD has internal protection)

Control Signals: - **Speed reference:** 0-10V analog, 4-20mA, or Modbus/RS485 - **Run/Stop:** Digital input (24V or dry contact) - **Fault output:** Relay output for E-stop circuit integration

Example VFD Wiring (2.2 kW spindle, Huanyang-style VFD): - Terminals R, S, T: 230V AC single-phase input (or 3-phase) - Terminals U, V, W: 3-phase output to spindle motor - Terminals DCM, VI: 0-10V speed reference from CNC controller - Terminals DCM, FOR: Run command (close to start spindle) - Terminals DCM, REV: Reverse command (rarely used for CNC)

End of Motor and Drive Sizing Appendix